

When the going gets hot, Agent 001 gets going

Agent 001 goes in search of ways to beat the heat

The phone rings, my mother screams, “beta, phone!”. Who can it be calling at this unholy hour (noon), when I’m on my vacation—I took a week off to play the latest game releases. It’s that guy I met a few weeks ago—the guy I talked into buying an AMD. “*Kya AMD Shay-em-dee*, this computer is *bekaar yaar*. I barely use it for *adha-ghanta* and it starts hanging”, he screams. Grabbing my umbrella and putting on my rubber chappals, I dash off to his place wondering, “How dare he think I gave him (Gasp!) bad advice. Agent 001 is *never* wrong!”

As soon as I entered his room, I realised what the problem was. My friend took his stinginess a little too far. As if he didn’t save enough money by taking my advice, he decided to be penny wise and pound foolish, and installed his AMD in a baby AT cabinet. The way he set it up, he could bake a cake in there. Now, he was faced with the choice of owning a 1.6 GHz frying pan, or spending (ouch!) on some cooling solutions.

We set out on our quest, and soon found ourselves in the dragons lair, the

infamous Lamington Road). The first vendor we came across blamed the problem on the AMD box pack. According to him the fans that are shipped with the CPUs are too small, and he offered me a bigger one, from some unheard-of company, for Rs 250. Respectfully declining, we made our way to the next piranha pond. Surprisingly he was well-informed. Between paan-spitting and whooping-cough breaks, he told us that cooling was an often overlooked and under-rated aspect of computing, buying bigger fans is just not enough.

He explained, in great detail, about how a cabinet should have good ventilation, and about the different types of heat sink-fan combinations available. Apparently, the cheaper ones are made of an aluminium alloy and have fan speeds of 4,000 rpm. Another type of combination is made up of copper embedded in aluminium, and costs around Rs 350 for normal brands, and Rs 750 for reputed brands such as Cooler master and Thermaltake. Even the rare Thermaltake series called Volcano are available, starting at Rs 800, although you might have to place an order and wait a few days for delivery.

The third type are entirely copper that offer the best heat conduction and therefore the best cooling. They are a little more expensive, starting at Rs 1,000, but are well worth the price. We also learnt that the fan’s speed, measured in rotations per minute, plays an important role—the cooling power is directly proportional to fan speeds. We were truly convinced, and even my parsimonious friend was willing to part with the dough for the good of his AMD. Unfortunately, the vendor was fresh out of stock.

A little dejected but armed with knowledge, we proceeded on our quest. We decided to try suburban areas such as Bandra and Andheri, and found that everyone was offering the Cooler Master, Thermaltake and ADDA combination, albeit at higher prices (Rs 50 to Rs 100 more than Lamington Road prices). We actually found a dealer selling a liquid cooling solution, but at a price that nearly gave my



ILLUSTRATION: Mahesh Benkar

friend a cardiac infarct—Rs 5,000 and above. We also learnt of the ‘38 degree cabinet’—it’s like a normal cabinet with a blower hole cut into the side panelling, just above where the CPU is placed, for better ventilation.

We finally decided on a Thermaltake Volcano 7 series fan with a copper base for Rs 650. Most people can make do with the box-packed heat sink-fan combinations, for everyday use, but overclockers and power users should use better cooling solutions. Also, we must remember that just a good cooling combo isn’t enough to secure top-notch performance, a well ventilated cabinet is a must—as my miserly friend learnt, the hard way.

After installing the fan and heat sink combination for my new best-buddy, I headed home, swelling with the knowledge that I had saved the world from itself, yet again. Dreaming of owning the liquid cooler (I made a mental note to put it down on my Christmas list for Santa), I headed back to my PC to see if I could beat my previous record of a non-stop 17 and a half hours of playing Postal 2. ■



■ Only buy what you need, most bundled cooling solutions are adequate for day-to-day work

■ Heat sinks are rated according to CPU speeds, so check whether the heat sink supports your CPU

■ Before buying a bigger heat sink, make sure it will fit in your cabinet

■ Big heat sink-fan combinations are heavy, thus they require additional retention mechanisms, such as screw’s. Check to see if your motherboard has the required pre-drilled holes near the CPU socket

■ Before you buy the heat-sink fan combination, check the under surface for the thermal compound, it is generally covered with a thin plastic sheet.

■ The right type of thermal compound must be used as specified by the CPU manufacturer. Check the type that your processor requires from its manufacturer’s Web site